Q1

Table 1 shows $\sigma_1 = 121.846$ (92%) of total energy
Figure 2, left side shows only after $k=5$, cumulative energy > 99%

So, lighting is low dimensional.

We know σ_4, σ_5, σ_6 quite big (at least 7% of energy)
Figure 2, right shows H_1 residuals ≈ 0.35

Why $k > 3$?

① N_1 — N_4 (4 light source directions) → shadows
② glasses & skin? → light glints (spike spots)
③ I dont know, but I suspect is head tilt position, Figure 3
because $N_1 - N_4$ should provide sample lighting for Projection (step 4)

Q2 Table 2 & Figure 2, right

	H_1	E_1
residuals (average)	0.36	0.15
reduction (percentage	10%	30%

Why behave different?

H_1: u_1, u_2, u_3, u_4 → lighting (Figure 1)
yeah, spatial allignment + shadows (Figure 3), should match more than E_3 smile

E_3: Figure 1, N_5 & N_6 got Front lighting
but cannot go below 13% because smile is geometric, non-rigid, Figure 3, right, shows "mustache shadow"

E_1 residual

As mentioned, my moustache, lips, cheeks (raised)

Implication

Face projection require low dimenston lighting subspace U_k ($k \simeq 3$ to 5)

pros: flexible to global lighting (dynamic)

cons: localised expressions (geometric) untracked

cons: spatial misallignment, very sensitive

Q3

Why measured = predicted (Table 1)?

economy SVD takes 6 σ ranks only

$$A - A_k = \sum_{i=k+1}^{6} \sigma_i u_i v_i^T \rightarrow \tilde{\Sigma}_k = \begin{pmatrix} 0 & & \\ & \sigma_{k+1} & \\ & & \sigma_6 \end{pmatrix}$$

$$\|A - A_k\|_F^2 = \sum_{i=k+1}^{6} \sigma_i^2$$

$$\|A\|_F^2 = \sum_{i=k+1}^{6} \sigma_i^2$$

Derivation

$$tr\left((U\tilde{\Sigma}_k V^T)^T (U\tilde{\Sigma}_k V^T)\right)$$

$$tr\left(V\tilde{\Sigma}_k U^T U \tilde{\Sigma}_k V^T\right) = \sum_{i=k+1}^{6} \sigma_i^2$$

- tr: Square (diagonal sum) $\sum_{i=k+1}^{6}$
- $U^T U$: I_6

Example, $k = 4$ (Table 1)

$$\sqrt{\frac{\sigma_5^2 + \sigma_6^2}{\sigma_1^2 + \sigma_2^2 + \cdots \sigma_6^2}} = 0.10647 \text{ (match)}$$

Why σ_1 5× larger?

① Baseline Face + Brightness (Figure 1, U_1 → my face emphasized)

② High similarity (linear?) cause same position + stationary